

Written Final Exam — Answer Key (Instructor Only)

Do not distribute to students. Version 1.1 · 50 questions · Pass = 35/50 (70%)

Multiple Choice

#	Ans	#	Ans	#	Ans	#	Ans
1	B	11	B	21	B	31	A
2	A	12	A	22	B	32	B
3	C	13	B	23	B	33	B
4	B	14	B	24	B	34	A
5	B	15	B	25	B	35	B
6	B	16	B	26	B	36	B
7	B	17	B	27	B	37	B
8	B	18	B	28	A	38	B
9	B	19	B	29	B	39	B
10	A	20	A	30	A	40	B

Short Answer — Accept equivalent correct wording

- V = I × R** (or $I = V/R$). **I = 36/12 = 3A**
- 1)** Verify complaint **2)** Gather information/history **3)** Visual & safety inspection **4)** Isolate systems **5)** Test components **6)** Perform repair **7)** Verify repair & road test
- Any three: safety glasses, face shield, Class 00 insulated gloves, leather over-gloves, apron, steel-toe boots
- Wrong year/model diagrams cause incorrect pinouts/colors/circuit paths and misdiagnosis
- Contacts are burned/pitted/high resistance — coil may still click but power delivery is poor
- Examples: LA needs watering/possible EQ vs Li no watering; chemistry-specific chargers; BMS vs SG testing; weight; cold charge rules — award point for two accurate contrasts
- Pedal up → key/enable on → then apply throttle (OEM-specific confirmation required)
- Any three: brakes, suspension, steering, drivetrain/transaxle, tires/wheels

9. Hydrogen gas from charging is explosive; ventilation reduces accumulation risk
10. **Diagnose before replace** (or equivalent clear statement)

Scoring notes

- MC: 1 point each (40)
- Short answer: 1 point each (10) — partial credit OK at instructor discretion (0.5)
- Critical safety wrong answers on related items should trigger coaching even if overall score passes